Dept. of Computer and Communication Engineering

Patuakhali Science and Technology University

7th Semester (Level-4, Semester-I), Midterm Examination of B.Sc. Engg. (CSE), January –June 2023

Course Code: CCE 415 Course Title: Network Routing and Switching Session 2019-2020

Credit Hour: 3.0 Full Marks: 15 Duration: 60 Minutes

1 a) How do you configure basic settings on a router to route between two directly connected networks, using CLI? Explain with a topology. 3

b) Write the exact answer with short explanation 4

i. When a computer is pinging another computer for the first time, which type of message does it send first to determine the MAC address of the other device?

ii. Write two functions of dynamic routing protocols.

iii. A network administrator has entered a static route to an Ethernet LAN that is connected to an adjacent router. However, the route is not shown in the routing table. Which command would the administrator use to verify that the exit interface is up?

iv. Consider the following routing table entry for R1: D 10.1.1.0/24 [90/2170112] via 209.165.200.226, 00:00:05, Serial0/0/0 What is the significance of the Serial0/0/0?

c) A network administrator is designing a small office network and is deciding whether to use static routing or a dynamic routing protocol. In which scenarios should the administrator use static routes instead of dynamic routing? What are the advantages and disadvantages of using static routes in these scenarios?

d) A network administrator has manually configured a static route on a router using the following command: Router(config)# ip route 192.168.2.0 255.255.255.0 192.168.1.2 .However, after applying the configuration, the administrator notices that the router is not forwarding traffic to the 192.168.2.0/24 network. What could be the possible reasons for this issue? Provide at least two explanations. 2

e) Assume a network administrator configures a static route on a router as follows using command: Router(config)# ip route 192.168.10.0 255.255.255.0 10.1.1.1 .Now the administrator wants to configure a backup route that will only be used if the primary route fails. (Consider administrative distance (AD) of a backup static route must be higher than AD of primary route.). How should the administrator configure a floating static route? What is the purpose of a floating static route, and how does it work? 3

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Final Examination of B. Sc. Engineering in CSE Level: 4 Semester: I Session: 2019-2020

Course Code	Course Title	January-June 2023	Credit: 03
CCE-415	Network Routing and Switching		Time: 03 Hr
			Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

[A.] Write the exact answer with short explanation 5

a. For packets to be sent to a remote destination, what three pieces of information must be configured on a host?

b. Consider the following routing table entry for R1: D 10.1.1.0/24 [90/2170112] via 10.2.1.1, 00:00:05, Serial0/0/0. What is the significance of the Serial0/0/0?

c. What command will enable a router to begin sending messages that allow it to configure a link-local address without using an IPv6 DHCP server?

d. What two tasks do dynamic routing protocols perform?

e. Having configured RIPv2 on an enterprise network, an engineer enters the command network 192.168.10.0 into router configuration mode. What is the result of entering this command?

f. Which two requirements are used to determine if a route can be considered an ultimate route in a router's routing table?

g. Which dynamic routing protocol was developed as an exterior gateway protocol to interconnect different Internet providers?

[B.] How does a router build a routing table of directly connected networks? 3

[C.] Illustrates the router packet-forwarding decision process with a flowchart. 3

[D.] Distinguish between unicast, multicast, and broadcast traffic in IPv6 with example. 3

[A.] "Routing protocols can be classified into different groups according to their characteristics" justify the statement. 4

[B.] Analyze the routes in the given output to determine the route source, AD and metric. 3

```
Gateway of last resort is not set

      10.0.0.0/16 is subnetted, 1 subnets
S        10.4.0.0 is directly connected, Serial0/0/0
      172.16.0.0/24 is subnetted, 3 subnets
C        172.16.1.0 is directly connected, FastEthernet0/0
C        172.16.2.0 is directly connected, Serial0/0/0
O        172.16.3.0 [90/2172416] via 172.16.2.1, 00:00:18, Serial0/0/0
C     192.168.1.0/24 is directly connected, Serial0/0/1
O  192.168.100.0/24 [110/65] via 172.16.2.1, 00:00:03, Serial0/0/0
O  192.168.110.0/24 [110/65] via 172.16.2.1, 00:00:03, Serial0/0/0
R  192.168.120.0/24 [120/1] via 172.16.2.1, 00:00:13, Serial0/0/0
```

[C.] Find the IPv4 address to its best route choice and explain why it is best choice? 3

172.17.128.0/17	10101100.00010001.1		
172.17.128.0/119	10101100.00010001.100		
172.17.160.0/19	10101100.00010001.101		
0.0.0.0/0			

172.17.218.83; 172.17.165.91; 172.17.124.36;172.17.169.20;172.17.155.116;172.16.12.140

[D.] Using the routing table below locate the network listed in the chart. Determine whether the networks are classified as Level 1, Level 1 parent, or Level 2 child routes. 4

```
Gateway of last resort is 0.0.0.0 to network 0.0.0.0

     192.0.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.0.2.0/30 is directly connected, Serial0/0/1
C       192.0.2.64/26 is directly connected, FastEthernet0/1
D    192.168.1.0/24 [90/2172416] via 192.168.2.1, 00:01:36, Serial0/0/0
C    192.168.2.0/24 is directly connected, Serial0/0/0
C    192.168.3.0/24 is directly connected, FastEthernet0/0
D    192.168.5.0/24[90/2172416] via 192.168.2.1, 00:01:36, Serial0/0/0
S*   0.0.0.0/0 is directly connected, Serial0/0/1
```

Specified Network	Route Type
0.0.0.0	
192.168.3.0/24	
192.0.02.64/26	
192.0.2.0/30	
192.0.2.24	

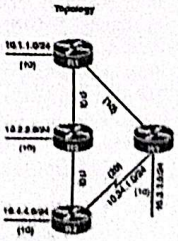
[A.] 2+3= 5

i) Draw the OSPF Shortest Path First (SPF) Tree of R1, and R4 router.

ii) Which three metric weights are set to zero by default when costs in EIGRP are being calculated?

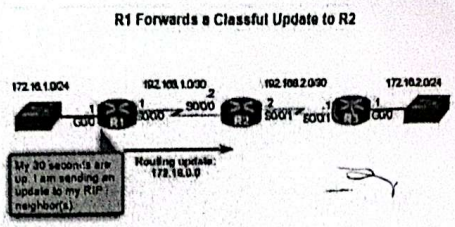
iii) By default, what is the OSPF cost for a link with Gigabit Ethernet interfaces?

iv) Which of the following attributes does the EIGRP topology table contain?



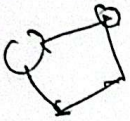
[B.] Write the process for forming OSPF Neighbor Adjacencies of R1 and R2 Router. 3

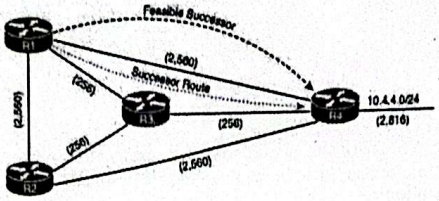
[C.] "Classful routing protocols also create problems in discontiguous networks. A discontiguous network is when subnets from the same classful major network address are separated by a different classful network address." Explain this shortcoming through the following topology.



3

[D.] Calculated the metrics from R1 and R2 for the 10.4.4.0/24 network through the following topology.





3

4 [A.] What is a Static Summary Route? Explained with topology and summary Static Route Calculation of that topology. 3

[B.] Assume your organization wants to block web traffic (HTTP) from any host on the 10.1.1.0/24 network to a specific web server with IP 192.168.100.100. Other types of traffic, like SSH or FTP, should still be allowed. What kind of ACL should be used, where should it be applied, and how should it be configured? Explain with suitable topology. 2

[C.] Assume you are creating an ACL to permit access from a specific subnet: 192.168.12.0/26.You know that to define this network in an ACL, you must use a wildcard mask instead of a subnet mask. What wildcard mask corresponds to the /26 subnet, and how would you write an ACL statement to permit traffic from this subnet? Provide the full breakdown of how the wildcard mask is calculated. 3

[D.] You are a network administrator for a company called TechVerse Inc., which is expanding its operations across three buildings — each with a dedicated router (R1 to R2 and R2 to R3). Each building hosts multiple departments (e.g., HR, Sales, IT), and you must segment them into VLANs for better security and traffic management. Each building/router must support at least 2 VLANs: VLAN 10 (192.168.10.0/24): HR, VLAN 20 (192.168.20.0/24): Sales and VLAN 30(192.168.30.0/24): IT. Use 192.168.100.0/30 subnets for links between routers. Assess: 6

i. Draw a network topology showing how the routers are connected and label all interfaces with IP addresses.
ii. Write command to create VLANs 10, 20, and 30 on one switch.
iii. Write command to configure trunk links between switches and router interfaces.
iv. Write command to configure sub-interfaces on R2 router for inter-VLAN routing
v. Write command to Verify VLANs and routing
vi. Why inter-VLAN routing is necessary?

5 [A.] What is an Access Control List (ACL)? What is the significance of the implicit "deny all" at the end of every ACL? Explain with example. 3

[B.] What happens if the MAC address table entry for a device times out and the switch receives a frame for that destination? 2

[C.] What is the primary function of DHCPv4 in a network? List the four steps of the DHCPv4 lease process. 3

[D.] Assume you are the network engineer for a mid-sized organization with four branch offices and the organization uses the 192.168.x.0/24 IP address range: Head Office (192.168.1.0/24), Sales Office (192.168.2.0/24), R&D Office (192.168.3.0/24), and Support Center (192.168.4.0/24). Each office has its own router, and all routers are interconnected through serial links. The connections between the routers are as follows: Head Office is connected to Sales Office and R&D Office, Sales Office is connected to Support Center, and R&D Office is also connected to Support Center. Use the 192.168.10.0/30 and 192.168.20.0/30 subnets for inter-router links. Assess : 6

i. Draw a network topology showing how the routers are connected and label all interfaces with IP addresses.
ii. Use next-hop static routes from Sales to R&D.
iii. Use a default static route on Support Center pointing to Sales Office.
iv. Configure a floating static route on Support Center to use R&D as a backup route to Head Office's network, with higher administrative distance.
v. On Head Office to Support Center, demonstrate a fully specified static route.
vi. Write the static route verification commands for one router based on the topology.

6 [A.] What is tagging and trunk in VLAN? Explain with diagram. 4

[B.] The store-and-forward method makes a forwarding decision on a frame after it has received the entire frame and checked the frame for errors. Which mechanism is used? Justify with proper figure and explanation. 3

[C.] Assume an organization wants to reduce broadcast traffic on its IPv6 network while still assigning DNS information dynamically. Which address assignment method should it choose—SLAAC, stateless DHCPv6, or stateful DHCPv6—and why? 2

[D.] You are a network administrator for a company. You are tasked with preventing the IP address 192.168.10.50 from accessing the entire 192.168.20.0/24 network. However, other users on the 192.168.10.0/24 network should retain access. You decide to use a standard ACL. Where would you place the ACL, and how would you configure it on a Cisco router? 2

[E.] What is a DHCP relay agent and why it is needed in some networks? Which command is used to configure a router interface to receive its IP address via DHCP and to configure DHCP relay agent? 3

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7

a) How is dissimilarity computed between objects described by nominal attributes?

b) How can we compute the dissimilarity between two binary attributes?

Consider the following transaction database. 8

Transaction ID	Items Bought
T1	{A, B, D}
T2	{B, C, E}
T3	{A, B, C, E}
T4	{B, C, D}
T5	{A, C, D, E}
T6	{A, B, C, D, E}

(a) Find all frequent itemsets using Apriori and FP-growth, respectively. Compare the efficiency of the two mining processes. Take, minimum support threshold = 2 and minimum confidence threshold = 50%.

(b) List all the *strong* association rules.

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Final Examination of B. Sc. Engineering in CSE Level: IV Semester: I Session: 2019-2020

Course Code	Course Title	January-June 2023	Credit: 03
CCE-417	Data Warehousing and Mining		Time: 03 Hr
			Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

1 [A.] Suppose that a group of 1500 people was surveyed. The gender of each person was noted. Each person was polled as to whether his or her preferred type of reading material was fiction or nonfiction. Thus, we have two attributes, *gender* and *preferred reading*. The observed frequency (or count) of each possible joint event is summarized in the contingency table shown in Table 1. 7

Table 1:2X2 Contingency Table Data

	male	*female*	*Total*
fiction	250	200	450
non fiction	50	1000	1050
Total	300	1200	1500

Are *gender* and *preferred reading* correlated? Note that for 1 degree of freedom, the χ^2 value needed to reject the hypothesis at the 0.001 significance level is 10.828.

[B.] Define the following terms with example 7

a) Metadata
b) Field Overloading
c) Uniqueness Rule
d) Consecutive Rule and
e) Null Rule

2 [A.] a) "Correlation does not imply causality"—justify the above statement. 2

b) Find out dissimilarity of diseases among Jack, Jim and Mary using the following patient record table (Table 2) which contains the attributes *name, gender, fever, cough, test-1, test-2, test-3*, and *test-4*, where *name* is an object identifier. 5

Table 2
Relational Table Where Patients Are Described by Binary Attributes

name	*gender*	*fever*	*cough*	*test-1*	*test-2*	*test-3*	*test-4*
Jack	M	Y	N	P	N	N	N
Jim	M	Y	Y	N	N	N	N
Mary	F	Y	N	P	N	P	N
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

[B.] Find out Cosine similarity among the documents of a self-prepared Term-Frequency Vector (Four documents will be in the document vector). 7

3 [A.] a) "Values have a meaningful order but magnitude between successive values is not known"—justify the above statement. 2

b) How we can compute the similarity between two binary attributes? Explain with suitable example. 5

[B.] a) What is five-number summary of a distribution? Explain with a suitable example using Boxplots. 4

b) How do you measure proximity for ordinal attributes? Describe with a suitable example. 3

4 [A.] Why is tree pruning useful in decision tree induction? What is a drawback of using a separate set of tuples to evaluate pruning? 2

[B.] The following contingency table (*Table 3*) summarizes supermarket transaction data, where *hot dogs* refers to the transactions containing hot dogs, *hot dogs* refers to the transactions that do not contain hot dogs, *hamburgers* refers to the transactions containing hamburgers, and *hamburgers* refers to the transactions that do not contain hamburgers. 12

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Table 3

	hot dogs	$\overline{\textit{hot dogs}}$	Σ_{row}
hamburgers	2000	500	2500
$\overline{\textit{hamburgers}}$	1000	1500	2500
Σ_{col}	3000	2000	5000

Ex. 6.14

a) Suppose that the association rule "*hot dogs*⇒*hamburgers*" is mined. Given a minimum support threshold of 25% and a minimum confidence threshold of 50%, is this association rule strong?

b) Based on the given data, is the purchase of *hot dogs* independent of the purchase of *hamburgers*? If not, what kind of *correlation* relationship exists between the two?

c) Compare the use of the *all_confidence*, *max_confidence*, *Kulczynski*, and *cosine* measures with *lift* and *correlation* on the given data.

5 [A.] Consider the following dataset (Table 4). Find the first stump using *AdaBoost* algorithm. Use *Gini Index* for internal calculation. 7

Table 4

Row No.	Gender	Age	Income	Illness
1	Male	41	40000	Yes
2	Male	54	30000	No
3	Female	42	25000	No
4	Female	40	60000	Yes
5	Male	46	50000	Yes

[B.] Suppose that we want to select between two prediction models, *M1* and *M2*. We have performed 10 rounds of 10-fold cross-validation on each model, where the same data partitioning in round *i* is used for both *M1* and *M2*. The error rates obtained for *M1* are 30.5, 32.2, 20.7, 20.6, 31.0, 41.0, 27.7, 26.0, 21.5, 26.0. The error rates for *M2* are 22.4,14.5, 22.4, 19.6, 20.7, 20.4, 22.1, 19.4, 16.2, 35.0. Comment on whether one model issignificantly better than the other considering a significance level of 1%. Ex. 8.14 5

[C.] Briefly outline the methods for addressing the class imbalance problem. Ex. 8.16 2

6 [A.] a) BIRCH applies a *multiphase* clustering technique. What are the primary phases? Pg. 465 5

b) Consider the numbers: 7, 10, 1, 19, 4, 17, 21, 31, 25, 28, 42. Take T=3. Construct the final B+ tree.

[B.] Give an application example where global outliers, contextual outliers, and collectiveoutliers are all interesting. What are the attributes, and what are the contextual andbehavioral attributes? How is the relationship among objects modeled in collectiveoutlier detection? Ex. 12.1 4

[C.] Prove that in DBSCAN, the density-connectedness is an equivalence relation. Ex 10.7 3

[D.] Briefly describe and give examples of each of the following approaches to clustering: *partitioning methods*, *hierarchical methods*, *density-based methods*, and *grid-based methods*. Ex. 10.1 2

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Patuakhali Science and Technology University

Department of Computer Science and Information Technology

Semester (Level-4, Semester-I), **Midterm Examination** of **B.Sc. Engg.(CSE)**, January-June/2023, Session: 2019-20

Course Code: CIT-411 Course Title: Compiler Design and Finite Automata

Full Marks: 15 Duration: 1 hour

1. Define syntax-directed translation. Illustrate a model of a compiler front end. Write the four components of *context-free grammar*. What is recursive-decent parsing? How do you eliminate the left-recursive from a grammar? 05
2. What is the function of the lexical analyzer? How does the lexical analyzer remove the white space and comments in a source program? 03
3. What is automata theory? List some of the applications of finite automata. Define decidability and intractability. 04
4. Prove that if $x \geq 4$, then $2^x \geq x^2$ by an induction on x, given the following BASIS. 03
   BASIS: IF $x = 4$, then 2^x and x^2 are both 16. Thus $2^4 \geq 4^2$ holds.

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Final Examination of B. Sc. Engineering in CSE Level: 4 Semester: I Session: 2019-2020

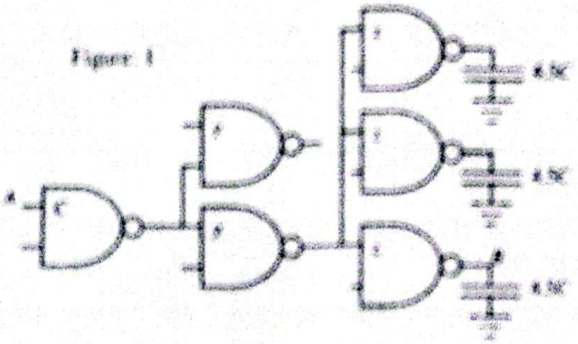
Course Code	Course Title	January-June 2023	Credit: 03
CCE-413	VLSI Design		Time: 03 Hr
			Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

1. [A.] Briefly explain the basic steps of CMOS IC fabrication process. 5

[B.] Optimize the circuit in Figure 1 to obtain the least delay along the path from A to B when the electrical effort of the path is 4.5. What is the least delay of this path, and how should the transistors be sized to achieve least (minimum) delay? (Consider logical, branching, and electrical efforts calculation along the path.) 4

Figure: 1



[C.] Consider an nMOS transistor in a 65 nm process with a minimum drawn channel length of 50 nm (Q = 25 nm). Let W/L = 4/2 Q (i.e., 0.1/0.05 Rm). In this process, the gate oxide thickness is 10.5 Å. Estimate the high field mobility of electrons to be 80 cm2/V · s at 70 °C. The threshold voltage is 0.3 V. Plot Ids vs. Vds for Vgs = 0, 0.2, 0.4, 0.6, 0.8, and 1.0 V using the long-channel model. 2

[D.] Sketch a stick and static diagram for a CMOS gate computing $Y = \overline{(A + B + C) \cdot D}$ and estimate the cell width and height. 3

2. [A.] Describe the different conditions for a transistor to be in cutoff, linear and saturation modes with suitable figure. 5

[B.] Implement the equation $F = \overline{(A + B)C + D + E} \cdot \overline{F} \cdot \overline{G}$ using complementary CMOS. Size the devices so that the output resistance is the same as that of an inverter with an NMOS W/L = 2 and PMOS W/L = 6. Which input pattern(s) would give the worst and best equivalent pull-up or pull-down resistance? 5

[C.] What is threshold voltage drop problem? Explain with example. 5

[D.] 2-input CMOS NAND and NOR gates have been designed with Rn = 1kΩ, Rp = 2kΩ, Cout = 8fF, and Cx = 2fF, where Cx is the node capacitance between the series transistors. 5

i. Calculate the worst-case rise and fall times for this NAND gate.
ii. Calculate the best-case rise time for this NAND gate.
iii. Calculate the worst-case rise time for a 2-input CMOS NOR gate.

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bias temp- Time dependent
Inst dielectric breakdown
BTI

3 [A.] What is pull up and pull down device? Sketch a 3-input NAND gate with transistor widths chosen to achieve effective rise and fall resistance equal to that of a unit inverter (R). Annotate the gate with its gate and diffusion capacitances. Estimate Elmore delay t_{pdf} and t_{pdr} for the 3-input NAND gate where the output is loaded with h identical NAND gates. 5

[B.] Show the Shockley first order formulas for computing the Current Ids, which flows through the transistor channel in linear and saturation modes. 3

[C.] Why branching is significant in delay calculation? Use appropriate diagram of inverter(s) and apply branching. 3

[D.] Sketch a static CMOS logic gate for the following functions: 3

$$\overline{(A+B)\cdot(C+D)\cdot(E+F+(C\cdot H))}$$

cost testing size

4 [A.] i. What are the differences between Hard and Soft failures in CMOS reliability? Provide examples. 4

ii. Define key challenges in predicting long-term reliability of CMOS devices. How can they be addressed? 3

[B.] i. Write the advantages & disadvantages of top-down nanofabrication techniques. 4

ii. What are the key advantages of using 4H-SiC over bulk silicon for high-temperature applications, and how does the bandgap of SiC contribute to these advantages? 3

5 [A.] i. Explain Monte Carlo simulations. How do they contribute to reliability testing in memory circuits? 5

ii. What is the key reliability challenges faced in advanced CMOS VLSI memory designs? 2

[B.] i. Discuss the key features of the fabrication process for high-temperature 4H-SiC CMOS technology. 4

ii. Describe the main advantages of using FPGAs over CPUs and GPUs. 3

6 [A.] i. Give an explanation of the advantages of using NMOS transistors in instrumentation amplifiers. 4

ii. What is the role of the CTM layer in MIM devices, and why is it used as a sacrificial layer? 3

[B.] i. What are the key reliability challenges faced in advanced CMOS VLSI memory designs? 4

ii. Explicate the main objective of design of a Novel CMOS Instrumentation Amplifier Using 90 nm Technology. Why is it important? 3

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Final Examination of B.Sc. Engineering in **(CSE) Level-4, Semester-I** Session: **2019-20**

Course Code	**Course Title**	**January-June** 2023	**Credit:** 03
CCE-411	**Algorithm Engineering**		**Time:** 03 Hr

Answer any 05 out of 06 Questions (Split answers are highly discouraged) **Marks:** 70

1 A. i. What is Map coloring? How can this term be used in different contexts? 3
ii. Describe the flow of Simple Genetic Algorithm (SGA) by illustrating it in a flowchart. 4

B. i. Explain heuristic search algorithm in artificial intelligence. What is the difference between algorithm and a heuristic? 4
ii. How to calculate time and space complexity from an algorithm in frequency count method? Use the following example to calculate. 3

```
Algorithm Multiply (P,Q,m)
{
for(inti=0;i<m;i++)
{
for (int j=0;j<m;j++)
        {
        R[i,j]=0;
        for(intk=0;k<m;k++)
                {
                R[i,j]=R[i,j]+ P[i,k]* Q[k,j];
                }}
} }
```

2 A. A table is provided that lists the height and weight of 12 individuals. 7

individual ID	Height	Weight
1	185	72
2	170	56
3	168	60
4	179	68
5	182	72
6	188	77
7	180	71
8	180	70
9	183	84
10	180	88
11	180	67
12	177	76

Utilize the K-means Clustering Algorithm to categorize the aforementioned data.

B. i. Explain Fitness Evaluation: Fitness Function / Objective Function in Genetic Algorithm. 3
ii. Determine the maximum and minimum values of $z = 6x + 4y$ under the following specified constraints by employing the Graphical Method. 4

$$x + 2y \leq 14$$
$$3x - y \geq 0$$
$$x - y \leq 2$$

3 A. i. What distinguishes genetic algorithms from traditional algorithms? 3
ii. Explain encoding technique: binary encoding in Genetic Algorithm. 2
iii. What are the types of linear programming? 2

Benefit
more advan
AI, mach;
Total point
Fast

3. B. The results of the engineering examination have been published. We are particularly focused on the courses pertaining to mathematics and computer science that are relevant to engineering students. Below is an example of the results for an engineering student. Student X has achieved scores of 6 in mathematics and 8 in computer-related courses. Your task is to determine whether X has passed or failed by employing K-Nearest Neighbor Classification, with K set to 3. 7

Student ID	Mathematics	Computer related courses	Result
1	4	3	fail
2	6	7	pass
3	7	8	pass
4	5	5	fail
5	8	8	pass

4. A Define asymptotic notation. What is the goal of asymptotic notation? Demonstrate and explain the asymptotic tight bound. 4

B What is complexity class? Compare NP-complete and NP-hard complexity classes with related examples. 3

C Formulate bin packing problem as an approximation problem. State the First-Fit 2-approximation algorithm for bin packing with an example. 4

D What is approximation ratio? Suppose you are familiar with vertex cover approximation algorithm and maximal matching. Prove the following statement: $|C| = 2 * |M| \leq 2*|C^*|$, where C is a vertex cover, M is a maximal matching, and C^* is an optimal cost. 3

5. A Define fixed parameter tractability. Formulate the k-vertex cover problem. Show that the exponent doesn't depend on k in the solution of the k-vertex cover. Design a fixed-parameter k-bounded search tree algorithm. How to analyze the complexity of a bounded search tree algorithm. 7

B What is the function of kernelization for designing an efficient algorithm? How to kernelize the giant problem? 4

C What is a heuristic technique? Mention the limitation of the heuristic technique for obtaining a solution. 3

6. A What are the problems with the deterministic algorithm? Demonstrate the block diagram of a randomized algorithm. Mention the advantages and difficulties of randomized algorithms. 4

B Distinguish between the Las Vegas algorithms and Monte Carlo algorithms with examples. Show the complexity of randomized Quick sort algorithm is $O(n \log n)$ using related example and parameters. 5

C What is minimum cut problem? State the contract algorithm with example. Describe the solution of minimum cut problem in a supposed graph using the contract algorithm. 5

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